SNS Lab 08

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Task 1 (Post lab)

~~Pre - Lab~~

Task - 1:

$$x(t) = \begin{cases} a\cos\left(\frac{\pi t}{2T}\right) & -T' < t < T' \\ 0 & T^3 < t < (T - T') \end{cases}$$

$$a_0 = \frac{1}{T}\int_{-T'}^{T} a\cos\left(\frac{\pi t}{2T'}\right)\cos(0)\,.dt$$

$$a_0 = \frac{1}{T}\int_{-T'}^{T'} a\cos\left(\frac{\pi t}{2T'}\right).dt$$

$$a_0 = \frac{a}{T}\int_{-T'}^{T'} \cos\left(\frac{\pi t}{2T'}\right).dt$$

$$a_0 = \frac{a}{T}\left[\frac{\sin\left(\frac{\pi}{2T'}\right)}{\pi/2T'}\right]_{-T'}^{T'}$$

$$a_0 = \frac{a}{T} \times \frac{2T'}{\pi}\left[\sin\left(\frac{\pi T'}{2T'}\right) - \sin\left(\frac{\pi(-T')}{2T'}\right)\right]$$

$$\boxed{a_0 = \frac{4aT'}{\pi T}}$$

$$a_k = \frac{2}{T}\int_{-T'}^{T'} a\cos\left(\frac{\pi T}{2T'}\right)\cos\left(\frac{k\pi t}{T}\right).dt$$

$$a_k = \frac{2a}{T}\int_{-T'}^{T'} \cos\left(\frac{\pi T}{2T'}\right)\cos\left(\frac{k\pi t}{T}\right).dt$$

$$a_k = \frac{2a}{2T}\int_{-T'}^{T'} \cos\left(\frac{\pi t}{2T'} - \frac{k\pi t}{T}\right) + \cos\left(\frac{\pi t}{2T'} + \frac{k\pi t}{T}\right).dt$$

$$a_k = \frac{a}{T}\int_{-T'}^{T'} \cos\left(\frac{T\pi t - 2T'k\pi t}{2TT'}\right) + \cos\left(\frac{T\pi t + 2T'k\pi t}{2TT'}\right).dt$$

$$a_k = \frac{a}{T}\left[\frac{\sin\left(\frac{T\pi - 2T'k\pi}{2TT'}\right)}{\frac{T\pi - 2T'k\pi}{2TT'}}\right]_{-T'}^{T'} + \left[\frac{\sin\left(\frac{T\pi + 2T'k\pi}{2TT'}\right)}{\frac{T\pi + 2T'k\pi}{2TT'}}\right]_{-T'}^{T'}$$

$$a_k = \frac{a}{T}\,2TT'\left(\frac{2}{T\pi - 2T'k\pi}\sin\left(\frac{T\pi - 2T'k\pi}{2T}\right) + \frac{2}{T\pi + 2T'k\pi}\sin\left(\frac{T\pi + 2T'k\pi}{2T}\right)\right)$$

$$a_k = 4aT'\left[\frac{\sin\left(\frac{T\pi - 2T'k\pi}{2T}\right)}{T\pi - 2T'k\pi} + \frac{\sin\left(\frac{T\pi + 2T'k\pi}{2T}\right)}{T\pi + 2T'k\pi}\right]$$

$$b_k = \frac{2}{T}\int_{-T'}^{T'} a\cos\left(\frac{\pi t}{2T'}\right)\sin\left(\frac{k\pi t}{T}\right) dt$$

$$= \frac{2a}{2T}\int_{-T'}^{T'} \sin\left(\frac{T\pi + 2T'k\pi}{2TT'}\right) - \sin\left(\frac{T\pi - 2T'k\pi}{2TT'}\right) dt$$

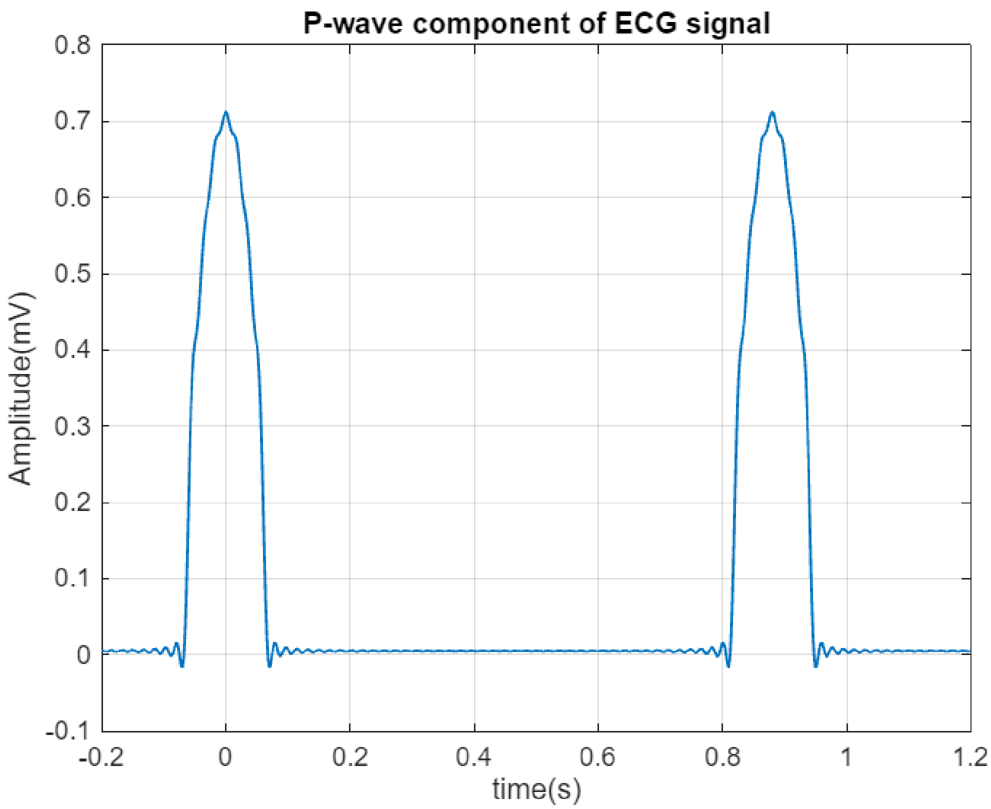
$$= \frac{2aTT'}{T}\left[\frac{1}{T\pi + 2T'k\pi} - \cos\left(\frac{T\pi + 2T'k\pi}{2TT'}\right) + \frac{\cos}{T\pi - 2T'k\pi}\left(\frac{T\pi - 2T'k\pi}{2TT'}\right)\right]_{-T'}^{T'}$$

$$b_k = a \times 2T'(0) = 0.$$

Task 2

```
%parameters
N=50;
a=0.35;
T=0.88;
Td=0.16;
t=-0.2:1/N/10:1.2;
k=1:N;

% Calculating a0 and ak
a0=(4*a*Td)/(pi*T);
ak=4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));

% Initializing xtNp with a0
bk=0;
xtNp=a0;

% Sum to compute xtNp
```

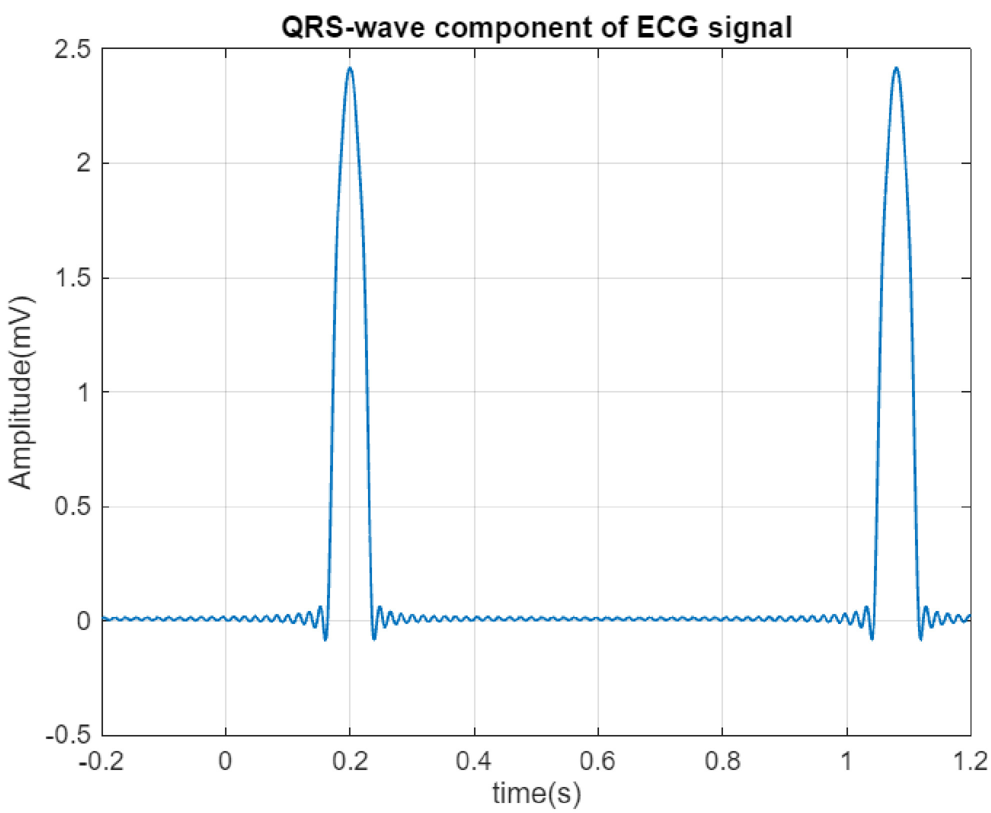
```
for kk=1:N
   xtNp=xtNp + ak(kk)*cos((2*kk*pi*t)/T);
end

% Ploting the P-wave component of ECG signal
figure;
plot(t, xtNp, 'linewidth', 1);
grid on;
xlabel('time(s)');
ylabel('Amplitude(mV)');
title('P-wave component of ECG signal');
```



Task 3

```
%parameters
N=50;
a=1.2;
T=0.88;
Td=0.08;
t0=0.2;
t=-0.2:1/N/10:1.2;
k=1:N;

% Calculating a0 and ak
a0 = (4*a*Td)/(pi*T);
ak=4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));
```

```
% Initializing xtNp with a0
bk=0;
xtNq = a0;

% Sum to compute xtNp
for kk=1:N
   xtNq = xtNq+ak(kk)*cos((2*kk*pi*(t-t0)/T));
end

% Ploting the QRS-wave component of ECG signal
figure;
plot(t, xtNq, 'linewidth',1);
grid on; xlabel('time(s)');
ylabel('Amplitude(mV)')
title('QRS-wave component of ECG signal');
```



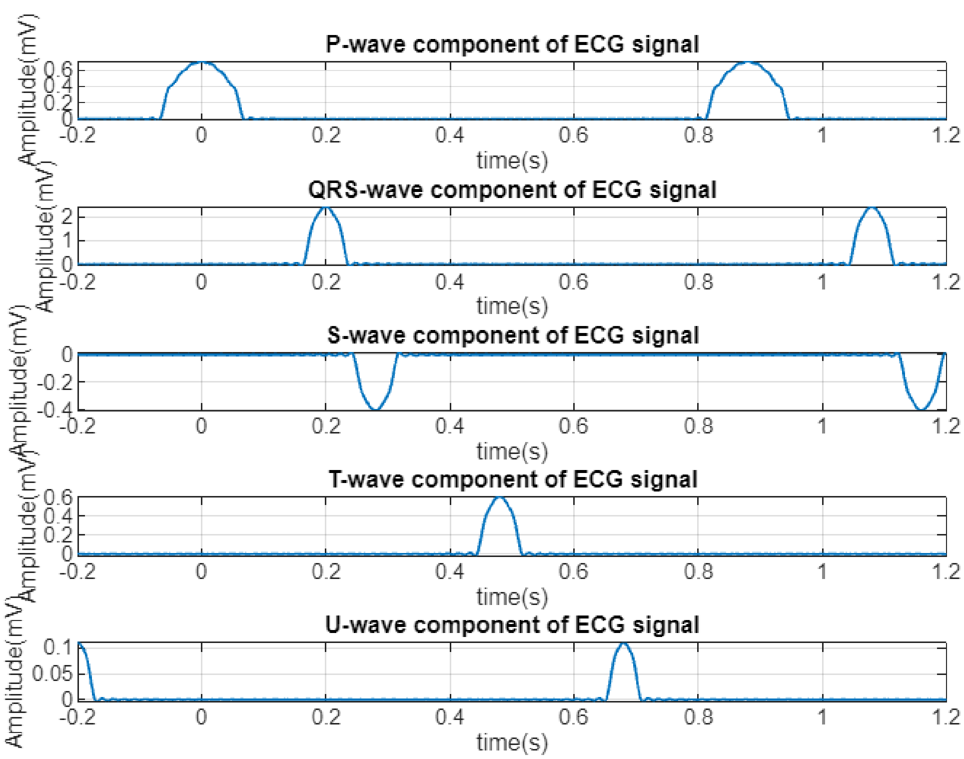
Task 4

```
%parameters Plot of P
N=50;
a=0.35;
T=0.88;
Td=0.16;
t=-0.2:1/N/10:1.2;
k=1:N;
```

```matlab
% Calculating a0 and ak
a0=(4*a*Td)/(pi*T);
ak=4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));

% Initializing xtNp with a0
xtNp=a0;

% Sum to compute xtNp
for kk=1:N
   xtNp=xtNp + ak(kk)*cos((2*kk*pi*t)/T);
end

% Ploting the P-wave component of ECG signal
subplot(5,1,1)
plot(t, xtNp, 'linewidth', 1);
grid on;
xlabel('time(s)');
ylabel('Amplitude(mV)');
title('P-wave component of ECG signal');

%parameters plot of QRS
N=50;
a=1.2;
T=0.88;
Td=0.08;
t0=0.2;
t=-0.2:1/N/10:1.2;
k=1:N;

% Calculating a0 and ak
a0 = (4*a*Td)/(pi*T);
ak=4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));

% Initializing xtNq with a0
xtNq = a0;

% Sum to compute xtNq
for kk=1:N
   xtNq = xtNq+ak(kk)*cos((2*kk*pi*(t-t0)/T));
end

% Ploting the QRS-wave component of ECG signal
subplot(5,1,2)
plot(t, xtNq, 'linewidth',1);
grid on; xlabel('time(s)');
ylabel('Amplitude(mV)')
title('QRS-wave component of ECG signal');
```

```matlab
%Parameters Plot of S
a = -0.2;
Td = 0.08;
t0 = 0.28;

% Calculating a0 and ak
a0 = (4*a*Td)/(pi*T);
ak = 4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));

% Initializing xtNp with a0
xtNs = a0;

% Sum to compute xtNp
for kk=1:N
xtNs = xtNs + ak(kk)*cos((2*kk*pi*(t-t0)/T));
end

% Ploting the S-wave component of ECG signal
subplot(5,1,3)
plot(t, xtNs, 'linewidth',1);
grid on; xlabel('time(s)');
ylabel('Amplitude(mV)')
title('S-wave component of ECG signal');

%Parameters Plot of T
a=0.3;
Td=0.08;
t0=0.48;

% Calculating a0 and ak
a0=(4*a*Td)/(pi*T);
ak=4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));

% Initializing xtNt with a0
xtNt = a0;

% Sum to compute xtNt
for kk=1:N
   xtNt = xtNt + ak(kk)*cos((2*kk*pi*(t-t0)/T));
end

% Ploting the T-wave component of ECG signal
subplot(5,1,4)
plot(t, xtNt, 'linewidth',1);
grid on; xlabel('time(s)');
ylabel('Amplitude(mV)')
title('T-wave component of ECG signal');
```

```
%Parameters Plot of U
a = 0.055;
Td = 0.06;
t0 = 0.68;

% Calculating a0 and ak
a0=(4*a*Td)/(pi*T);
ak=4*a*Td*((sin((pi*T-2*k*pi*Td)./2*T)./(pi*T-2*k*pi*Td))+(sin((pi*T+2*k*pi*Td)./
2*T)./(pi*T+2*k*pi*Td)));

% Initializing xtNu with a0
xtNu = a0;

% Sum to compute xtNu
for kk=1:N
   xtNu = xtNu + ak(kk)*cos((2*kk*pi*(t-t0)/T));
end

% Ploting the U-wave component of ECG signal
subplot(5,1,5)
plot(t, xtNu, 'linewidth',1);
grid on; xlabel('time(s)');
ylabel('Amplitude(mV)')
title('U-wave component of ECG signal');
```



Task 5

```
ECG = xtNp + xtNq + xtNs + xtNt + xtNu;
figure; plot(t,ECG, 'linewidth',1)
title ("ECG Wave");
grid on;
```